

Multi-variable example:

A researcher is studying factors that influence employee productivity (measured in units produced per hour) at a manufacturing plant. They collect data from 50 employees and run a multiple regression with the following independent variables:

- Years of Experience (continuous variable, ranging from 0-20 years)
- Training Hours (continuous variable, hours of training received in past year)
- Shift (categorical: Day shift = 0, Night shift = 1)

Regression Output:

Dependent Variable: Productivity (units/hour)

Sample size: $n = 50$

Coefficient Table:

Variable	Coefficient (b)	Standard Error	t-statistic	p-value
Intercept	12.5	1.8	6.94	<0.001
Years of Experience	0.85	0.25	3.40	0.001
Training Hours	0.45	0.15	3.00	0.004
Night Shift	-2.3	0.80	-2.88	0.006

Model Summary:

- $R^2 = 0.62$
- Adjusted $R^2 = 0.59$
- F-statistic = 24.8, $p < 0.001$
- Standard Error of the Estimate = 3.2

Correlation Matrix:

	Productivity	Experience	Training	Shift
Productivity	1.00	0.68	0.52	-0.38
Experience	0.68	1.00	0.15	0.08
Training	0.52	0.15	1.00	-0.12
Shift	-0.38	0.08	-0.12	1.00

Questions:

- a) Interpret each regression coefficient in practical terms.

- b) Is the overall model statistically significant? What does this tell us?

- c) Which predictors are statistically significant at the $\alpha = 0.05$ level?
- d) Predict the productivity for a day shift employee with 10 years of experience and 40 training hours.
- e) Interpret the R^2 value. What does it tell us about the model's explanatory power?
- f) Why might the Adjusted R^2 be lower than R^2 ? When is this distinction important?
- g) Compare the relative importance of Years of Experience vs. Training Hours. Which appears to have a stronger effect on productivity?
- h) Are there any concerns about multicollinearity based on the correlation matrix?

- i) What does the negative coefficient for Night Shift suggest? Provide a practical interpretation.
- j) If the company wanted to increase productivity most cost-effectively, what would you recommend based on these results?